

ENV 790.30 - Time Series Analysis for Energy Data | Spring 2026

Assignment 3 - Second Version

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., 'LuanaLima_TSA_A03_Sp25.Rmd'). Then change 'Student Name' on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

Please keep this R code chunk options for the report. It is easier for us to grade when we can see code and output together. And the tidy.opts will make sure that line breaks on your code chunks are automatically added for better visualization.

```
library(cowplot)
library(forecast)
library(readxl)
library(tidyverse)
library(tseries)
```

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

Questions

Consider the same data you used for A2 from the spreadsheet. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. This time you will work only with the following columns: **Total Renewable Energy Production**; and **Hydroelectric Power Consumption**.

Create a data frame structure with these two time series only.

R packages needed for this assignment: 'forecast', 'tseries', and 'Kendall'. Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
outlook <- read_excel("./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx",
  sheet = "Monthly Data",
  skip = 10)
outlook <- outlook[-c(1),
  ]
```

```

outlook <- outlook %>%
  select("Month", "Total Renewable Energy Production",
         "Hydroelectric Power Consumption")
outlook <- outlook %>%
  mutate(across(c("Total Renewable Energy Production",
                  "Hydroelectric Power Consumption"),
               as.numeric))
history <- ts(outlook[,
  -c(1)], start = c(1973,
  1), frequency = 12)

```

##Trend Component

Q1

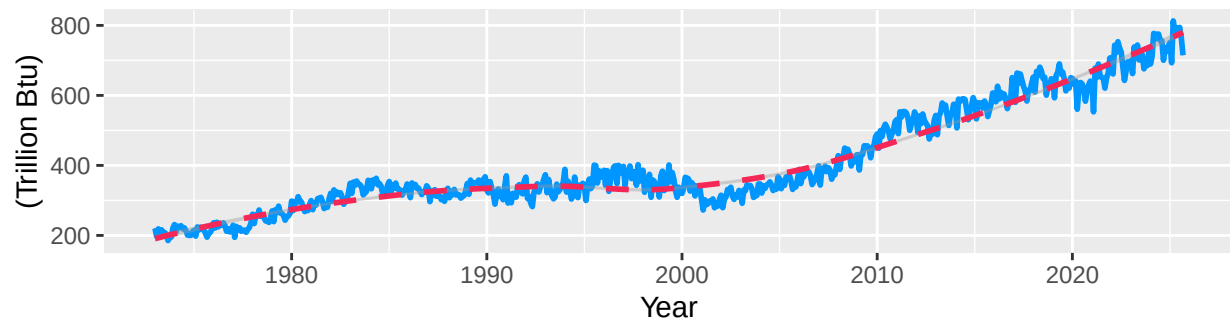
For each series (Total Renewable Production and Hydroelectric Consumption) create three plots arranged in a row (side-by-side): (1) time series plot, (2) ACF, (3) PACF. Use `cowplot::plot_grid()` to place them in a grid.

```

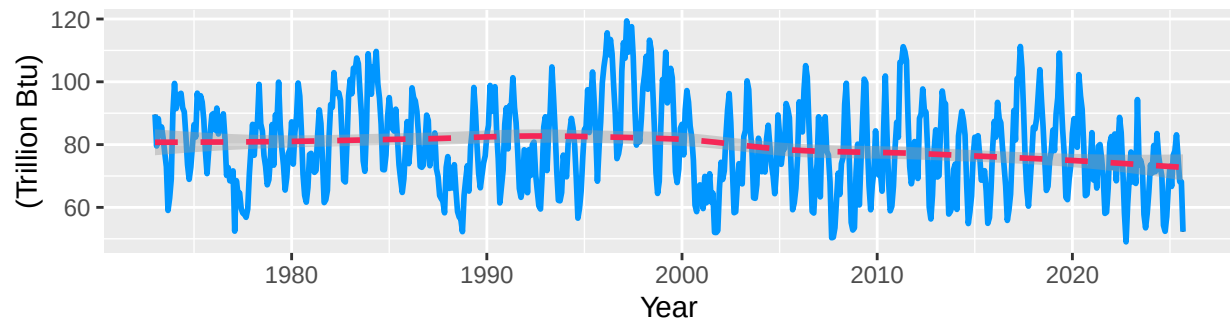
p1 <- autoplot(history[,
  1]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
    linewidth = 1,
    linetype = "dashed") +
  theme_update() +
  labs(title = "Time Series For Total Renewable Energy Production",
    subtitle = "From 1973 To 2025",
    x = "Year", y = "(Trillion Btu)")
p2 <- autoplot(history[,
  2]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
    linewidth = 1,
    linetype = "dashed") +
  theme_update() +
  labs(title = "Time Series For Hydroelectric Power Consumption",
    subtitle = "From 1973 To 2025",
    x = "Year", y = "(Trillion Btu)")
summary_first <- plot_grid(p1,
  p2, ncol = 1)
summary_first

```

Time Series For Total Renewable Energy Production
From 1973 To 2025

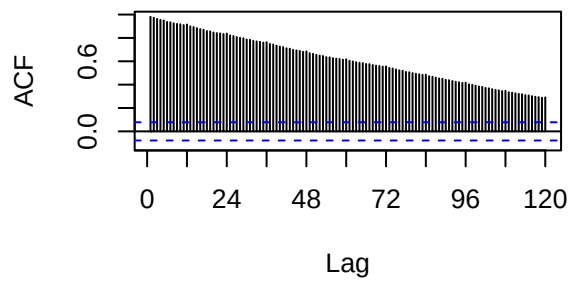


Time Series For Hydroelectric Power Consumption
From 1973 To 2025

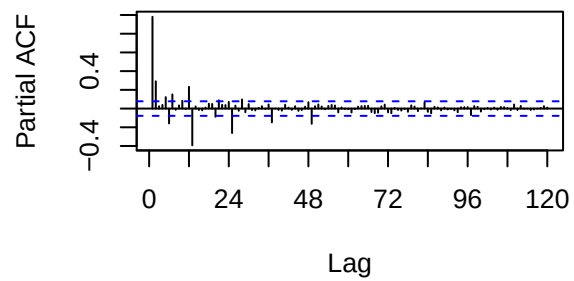


```
par(mfrow = c(2, 2))
Acf(history[, 1], lag.max = 120,
    type = "correlation",
    plot = TRUE, main = paste("Total Renewable Energy Production"))
Pacf(history[, 1], lag.max = 120,
    plot = TRUE, main = paste("Total Renewable Energy Production"))
Acf(history[, 2], lag.max = 120,
    type = "correlation",
    plot = TRUE, main = paste("Hydroelectric Power Consumption"))
Pacf(history[, 2], lag.max = 120,
    plot = TRUE, main = paste("Hydroelectric Power Consumption"))
```

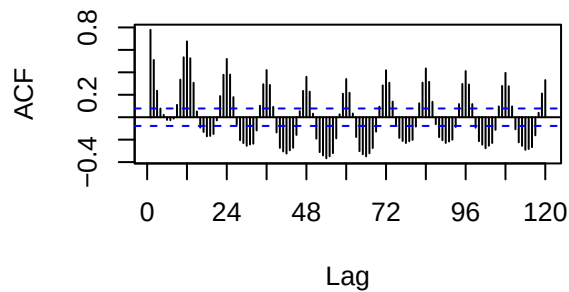
Total Renewable Energy Production



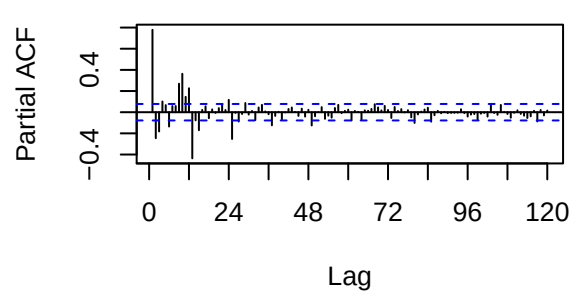
Total Renewable Energy Production



Hydroelectric Power Consumption



Hydroelectric Power Consumption



Q2

From the plot in Q1, do the series Total Renewable Energy Production and Hydroelectric Power Consumption appear to have a trend? If yes, what kind of trend?

Answer: Total renewable energy production had a significant upward tendency. The downward tendency in hydroelectric power consumption was negligible.

Q3

Use the `lm()` function to fit a linear trend to the two time series. Ask R to print the summary of the regression. Interpret the regression output, i.e., slope and intercept. Save the regression coefficients for further analysis.

```
t <- c(1:633)
renewable_first_model <- lm(history[,
  1] ~ t)
summary(renewable_first_model)
```

```
##
## Call:
## lm(formula = history[, 1] ~ t)
##
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -154.81  -39.55   12.52   41.49  171.15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.44868    5.11085   33.55  <2e-16 ***
## t           0.74999    0.01397   53.69  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.22 on 631 degrees of freedom
## Multiple R-squared:  0.8204, Adjusted R-squared:  0.8201
## F-statistic: 2883 on 1 and 631 DF, p-value: < 2.2e-16
```

```
beta0_renewable_first <- as.numeric(renewable_first_model$coefficients[1])
beta1_renewable_first <- as.numeric(renewable_first_model$coefficients[2])
hydro_first_model <- lm(history[,
  2] ~ t)
summary(hydro_first_model)
```

```
##
## Call:
## lm(formula = history[, 2] ~ t)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -30.190 -10.214  -0.715   8.909  39.723
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 83.223802    1.110552  74.939  < 2e-16 ***
## t          -0.012199    0.003035  -4.019 6.55e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.95 on 631 degrees of freedom
## Multiple R-squared:  0.02496, Adjusted R-squared:  0.02342
## F-statistic: 16.15 on 1 and 631 DF, p-value: 6.547e-05
```

```
beta0_hydro_first <- as.numeric(hydro_first_model$coefficients[1])
beta1_hydro_first <- as.numeric(hydro_first_model$coefficients[2])
```

Answer: In December 1972, the total renewable energy production was theoretically 171.44868 Trillion Btu. If one month passes, the total renewable energy production might increase by 0.74999 Trillion Btu. Given the very small p-value for coefficients and F-statistic, the linear model might be considerable.

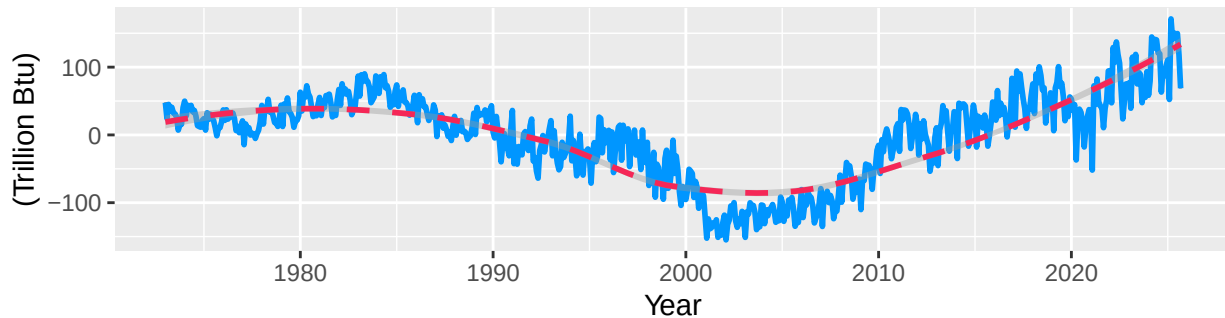
Answer: In December 1972, the hydroelectric power consumption was theoretically 83.223802 Trillion Btu. If one month passes, the hydroelectric power consumption might decrease by 0.012199 Trillion Btu. Given the very small p-value for coefficients and F-statistic, the linear model might be also considerable. However, since the adjusted R-squared is only 0.02342, the accuracy of model might be unfavorable.

Q4

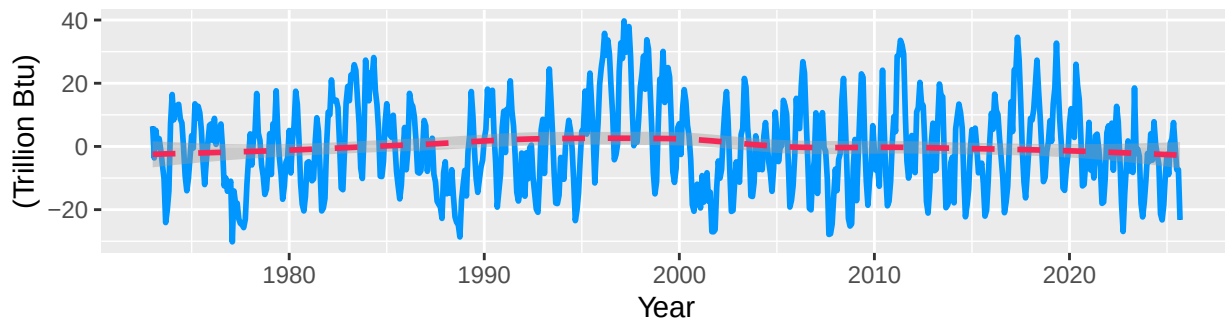
Use the regression coefficients to detrend each series (subtract fitted linear trend). Plot detrended series and compare with the original time series from Q1. Describe what changed.

```
detrend_outlook <- outlook %>%
  mutate(r1 = `Total Renewable Energy Production` -
    (beta0_renewable_first +
      beta1_renewable_first *
        t), h1 = `Hydroelectric Power Consumption` -
    (beta0_hydro_first +
      beta1_hydro_first *
        t))
detrend_history <- ts(detrend_outlook[,
  -c(1)], start = c(1973,
  1), frequency = 12)
p3 <- autoplot(detrend_history[,
  3]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
    linewidth = 1,
    linetype = "dashed") +
  theme_update() +
  labs(title = "Detrended Time Series For Total Renewable Energy Production",
    subtitle = "From 1973 To 2025",
    x = "Year", y = "(Trillion Btu)")
p4 <- autoplot(detrend_history[,
  4]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
    linewidth = 1,
    linetype = "dashed") +
  theme_update() +
  labs(title = "Detrended Time Series For Hydroelectric Power Consumption",
    subtitle = "From 1973 To 2025",
    x = "Year", y = "(Trillion Btu)")
summary_second <- plot_grid(p3,
  p4, ncol = 1)
summary_second
```

Detrended Time Series For Total Renewable Energy Production
From 1973 To 2025



Detrended Time Series For Hydroelectric Power Consumption
From 1973 To 2025



Answer: For total renewable energy production, there is a quadratic trend. The minimum approximately occurs at 2003.

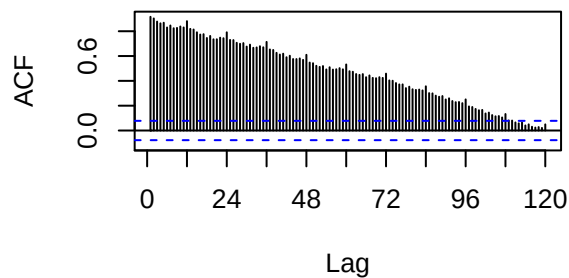
Answer: For hydroelectric power consumption, any obvious trends cannot be detected.

Q5

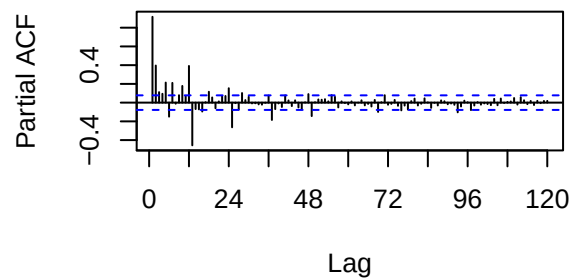
Plot ACF and PACF for the detrended series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side to make it easier to compare. Did the plots change? How?

```
par(mfrow = c(2, 2))
Acf(detrend_history[,
  3], lag.max = 120,
  type = "correlation",
  plot = TRUE, main = paste("Total Renewable Energy Production"))
Pacf(detrend_history[,
  3], lag.max = 120,
  plot = TRUE, main = paste("Total Renewable Energy Production"))
Acf(detrend_history[,
  4], lag.max = 120,
  type = "correlation",
  plot = TRUE, main = paste("Hydroelectric Power Consumption"))
Pacf(detrend_history[,
  4], lag.max = 120,
  plot = TRUE, main = paste("Hydroelectric Power Consumption"))
```

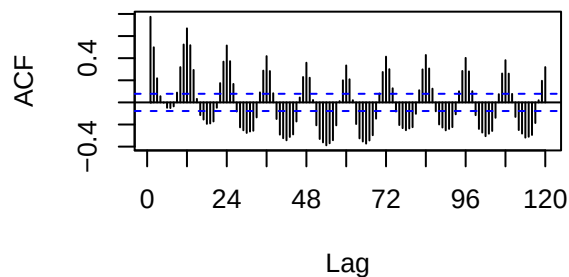
Total Renewable Energy Production



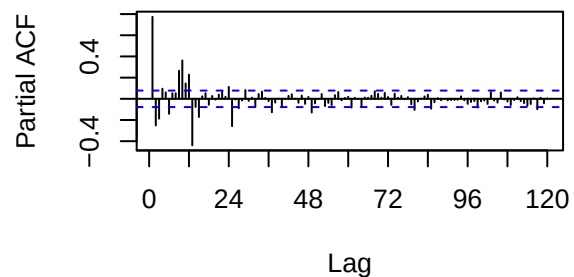
Total Renewable Energy Production



Hydroelectric Power Consumption



Hydroelectric Power Consumption



Answer: PACF plots of both series did not changed apparently. However, for total renewable energy production, ACF reached zero with lag 120. It might be a breakthrough of overcoming the trend.

Seasonal Component

Set aside the detrended series and consider the original series again from Q1 to answer Q6 to Q8.

Q6

Just by looking at the time series and the acf plots, do the series seem to have a seasonal trend? No need to run any code to answer your question. Just type in you answer below.

Answer: Total renewable energy production may not have a seasonal trend. However, hydroelectric power consumption seems to have a seasonal trend.

Q7

Use function `lm()` to fit a seasonal means model (i.e. using the seasonal dummies) to the two time series. Ask R to print the summary of the regression. Interpret the regression output. From the results, which series have a seasonal trend? Do the results match you answer to Q6?


```
dummy_renewable <- seasonaldummy(history[,
1])
renewable_second_model <- lm(history[,
1] ~ dummy_renewable)
summary(renewable_second_model)
```

```
##
## Call:
## lm(formula = history[, 1] ~ dummy_renewable)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -213.33  -97.36  -59.88   121.55   389.62
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      417.265      21.096   19.779 <2e-16 ***
## dummy_renewableJan      2.090      29.693    0.070  0.944
## dummy_renewableFeb    -34.524      29.693   -1.163  0.245
## dummy_renewableMar      5.956      29.693    0.201  0.841
## dummy_renewableApr     -6.900      29.693   -0.232  0.816
## dummy_renewableMay      8.162      29.693    0.275  0.784
## dummy_renewableJun     -2.231      29.693   -0.075  0.940
## dummy_renewableJul      3.864      29.693    0.130  0.897
## dummy_renewableAug     -3.978      29.693   -0.134  0.893
## dummy_renewableSep    -29.033      29.693   -0.978  0.329
## dummy_renewableOct    -19.937      29.834   -0.668  0.504
## dummy_renewableNov    -20.617      29.834   -0.691  0.490
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 152.1 on 621 degrees of freedom
## Multiple R-squared:  0.008243, Adjusted R-squared: -0.009324
## F-statistic: 0.4692 on 11 and 621 DF, p-value: 0.9223
```

```
beta0_renewable_second <- as.numeric(renewable_second_model$coefficients[1])
beta1_renewable_second <- as.numeric(renewable_second_model$coefficients[2:12])
dummy_hydro <- seasonaldummy(history[,
2])
hydro_second_model <- lm(history[,
2] ~ dummy_hydro)
summary(hydro_second_model)
```

```
##
## Call:
## lm(formula = history[, 2] ~ dummy_hydro)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.895  -6.368  -0.595    6.213   32.557
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      79.724      1.436  55.511 < 2e-16 ***
## dummy_hydroJan    4.951      2.021   2.449 0.014591 *
## dummy_hydroFeb   -2.415      2.021  -1.195 0.232608
## dummy_hydroMar    7.116      2.021   3.520 0.000463 ***
## dummy_hydroApr    5.614      2.021   2.777 0.005649 **
## dummy_hydroMay   14.080      2.021   6.965 8.38e-12 ***
## dummy_hydroJun   10.780      2.021   5.333 1.36e-07 ***
## dummy_hydroJul    4.003      2.021   1.980 0.048091 *
## dummy_hydroAug   -5.320      2.021  -2.632 0.008710 **
## dummy_hydroSep  -16.598      2.021  -8.211 1.28e-15 ***
## dummy_hydroOct  -16.329      2.031  -8.040 4.56e-15 ***
## dummy_hydroNov  -10.782      2.031  -5.308 1.54e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.36 on 621 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4621
## F-statistic: 50.35 on 11 and 621 DF,  p-value: < 2.2e-16
```

```
beta0_hydro_second <- as.numeric(hydro_second_model$coefficients[1])
beta1_hydro_second <- as.numeric(hydro_second_model$coefficients[2:12])
```

Answer: For total renewable energy production, the very large p-value does not indicate seasonality.

Answer: Given the small p-value for coefficients and F-statistic, hydroelectric power consumption may have a seasonal trend. In addition, hydroelectric power consumption is prominent in May and June. Hydroelectric power consumption is modest in September, October, and November. My assumption is correct.

Q8

Use the regression coefficients from Q7 to deseason the series. Plot the deseason series and compare with the plots from part Q1. Did anything change?

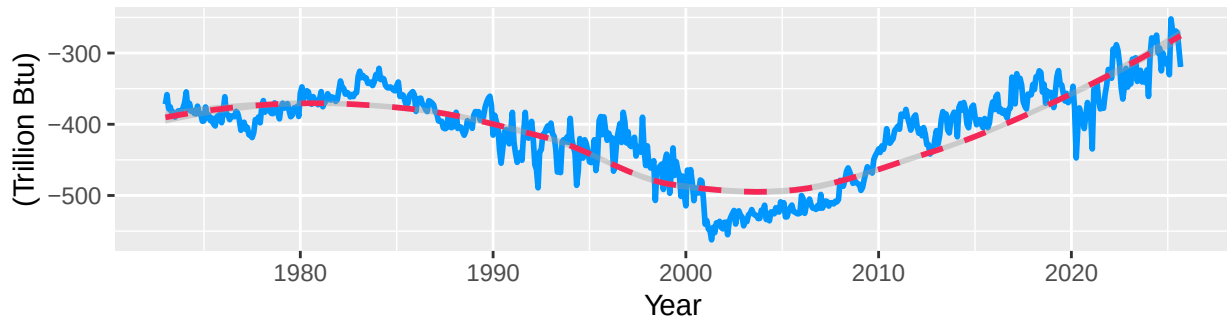
```
component_renewable <- array(0,
  633)
for (i in 1:633) {
  component_renewable[i] <- beta0_renewable_second +
    beta1_renewable_second %*%
    dummy_renewable[i,
  ]
}
component_hydro <- array(0,
  633)
for (i in 1:633) {
  component_hydro[i] <- beta0_hydro_second +
    beta1_hydro_second %*%
    dummy_hydro[i,
  ]
}
```

```

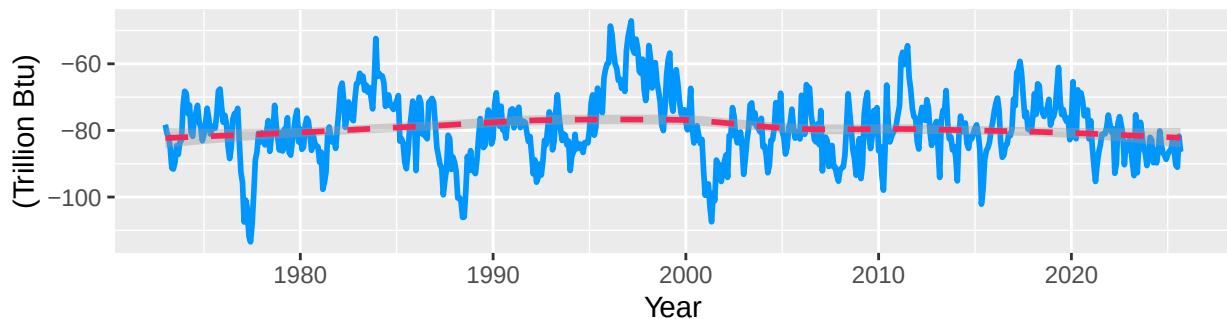
deseason_outlook <- detrend_outlook %>%
  mutate(r2 = r1 -
    component_renewable,
    h2 = h1 - component_hydro)
deseason_history <- ts(deseason_outlook[,
  -c(1)], start = c(1973,
  1), frequency = 12)
p5 <- autoplot(deseason_history[,
  5]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
  linewidth = 1,
  linetype = "dashed") +
  theme_update() +
  labs(title = "Deseasoned Time Series For Total Renewable Energy Production",
  subtitle = "From 1973 To 2025",
  x = "Year", y = "(Trillion Btu)")
p6 <- autoplot(deseason_history[,
  6]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
  linewidth = 1,
  linetype = "dashed") +
  theme_update() +
  labs(title = "Deseasoned Time Series For Hydroelectric Power Consumption",
  subtitle = "From 1973 To 2025",
  x = "Year", y = "(Trillion Btu)")
summary_third <- plot_grid(p5,
  p6, ncol = 1)
summary_third

```

Deseasoned Time Series For Total Renewable Energy Production
From 1973 To 2025



Deseasoned Time Series For Hydroelectric Power Consumption
From 1973 To 2025



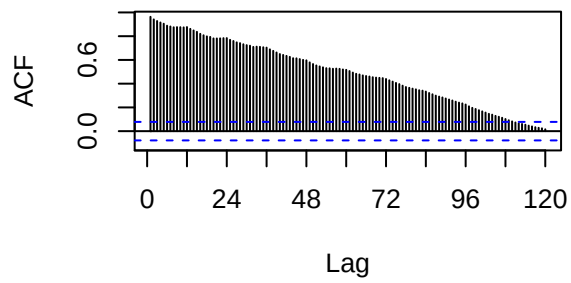
Answer: All values of energy have become negative. The total renewable energy production still has a quadratic trend. Moreover, the distribution of hydroelectric power consumption seems to become more random but concentrated.

Q9

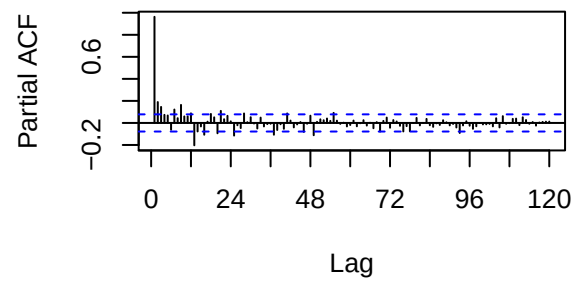
Plot ACF and PACF for the deseason series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side. Did the plots change? How?

```
par(mfrow = c(2, 2))
Acf(deseason_history[,
  5], lag.max = 120,
  type = "correlation",
  plot = TRUE, main = paste("Total Renewable Energy Production"))
Pacf(deseason_history[,
  5], lag.max = 120,
  plot = TRUE, main = paste("Total Renewable Energy Production"))
Acf(deseason_history[,
  6], lag.max = 120,
  type = "correlation",
  plot = TRUE, main = paste("Hydroelectric Power Consumption"))
Pacf(deseason_history[,
  6], lag.max = 120,
  plot = TRUE, main = paste("Hydroelectric Power Consumption"))
```

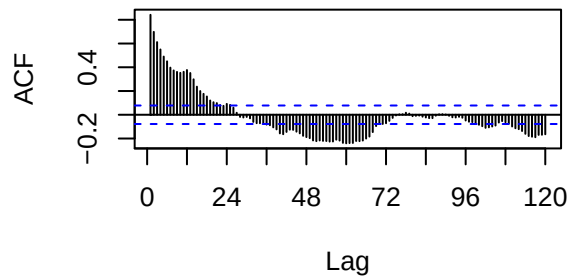
Total Renewable Energy Production



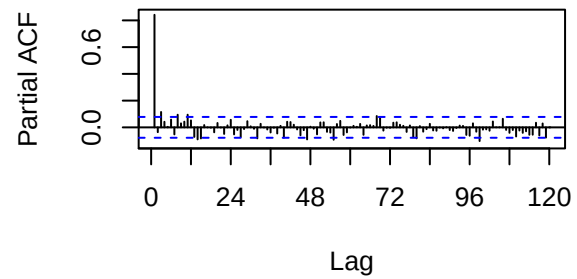
Total Renewable Energy Production



Hydroelectric Power Consumption



Hydroelectric Power Consumption



Answer: The most prominent change applies to ACF of hydroelectric power consumption. The reduction of fluctuating patterns may indicate that the removal of seasonal trend is effective.